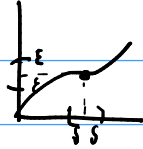


$$\sqrt{x^2 - x} \quad x(x-1) \geq 0 \quad x \leq 0 \vee x \geq 1 \quad \lim_{x \rightarrow 0} \sqrt{x^2 - x} \text{ NON ESISTE}$$

PERCHÉ $\lim_{x \rightarrow 0^+} \sqrt{x^2 - x}$ NON ESISTE

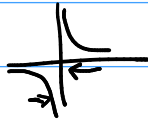
$A \subseteq \mathbb{R}$, $A \neq \emptyset$, x_0 È PUNTO DI ACCUMULAZIONE $\Leftrightarrow \forall U$ INTORNO DI x_0 $(A \cap U) \setminus \{x_0\} \neq \emptyset$

• $x_0 \in \mathbb{R}$, $l \in \mathbb{R}$



$\forall \varepsilon > 0 \exists \delta > 0$ t.c. $|f(x) - l| < \varepsilon \quad \forall x$ t.c. $0 < |x - x_0| < \delta$

• $x_0 \in \mathbb{R}$, $l = +\infty$

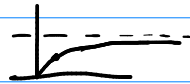


$\forall M \in \mathbb{R} \exists \delta > 0$ t.c. $f(x) > M \quad \forall x$ t.c. $0 < |x - x_0| < \delta$

• $x_0 \in \mathbb{R}$, $l = -\infty$

$\exists M \in \mathbb{R} \exists \delta > 0$ t.c. $f(x) < M \quad \forall x$ t.c. $0 < |x - x_0| < \delta$

• $x_0 = +\infty$, $l \in \mathbb{R}$



$\forall \varepsilon > 0, \exists a \in \mathbb{R}$ t.c. $|f(x) - l| < \varepsilon \quad \forall x > a$

• $x_0 = -\infty$, $l \in \mathbb{R}$

$\forall \varepsilon > 0, \exists a \in \mathbb{R}$ t.c. $|f(x) - l| < \varepsilon \quad \forall x < a$

• $x_0 = +\infty$, $l = +\infty$



$\forall M \in \mathbb{R}, \exists a \in \mathbb{R}$ t.c. $f(x) > M \quad \forall x > a$

• $x_0 = +\infty$, $l = -\infty$ $\forall M \in \mathbb{R}, \exists a \in \mathbb{R}$ t.c. $f(x) < M \quad \forall x > a$

• $x_0 = -\infty$, $l = -\infty$ $\forall M \in \mathbb{R}, \exists a \in \mathbb{R}$ t.c. $f(x) < M \quad \forall x < a$

• $x_0 = -\infty$, $l = +\infty$ $\forall M \in \mathbb{R}, \exists a \in \mathbb{R}$ t.c. $f(x) > M \quad \forall x < a$

$\lim_{x \rightarrow 1} f(x) = 3 \quad \forall \varepsilon > 0, \exists \delta > 0$ t.c. $|f(x) - 3| < \varepsilon \quad \forall x$ t.c. $0 < |x - 1| < \delta$

$\lim_{x \rightarrow -\infty} f(x) = -4 \quad \forall \varepsilon > 0, \exists a \in \mathbb{R}$ t.c. $|f(x) + 4| < \varepsilon \quad \forall x < a$

$\lim_{x \rightarrow \pi} f(x) = +\infty \quad \forall M \in \mathbb{R}, \exists \delta > 0$ t.c. $f(x) > M \quad \forall x$ t.c. $0 < |x - \pi| < \delta$

VERIFICARE: • $\lim_{x \rightarrow 0} x^2 = 0 \quad \forall \varepsilon > 0, \exists \delta > 0$ t.c. $|x^2| < \varepsilon \quad \forall x$ t.c. $0 < |x| < \delta$

È VERO PERCHÉ: $|x^2| < \varepsilon \Leftrightarrow x^2 < \varepsilon \Leftrightarrow |x| < \sqrt{\varepsilon} \Rightarrow 0 < |x| < \sqrt{\varepsilon} \Rightarrow \delta = \sqrt{\varepsilon}$

$$\bullet \lim_{x \rightarrow +\infty} \frac{x^2}{x+1} = +\infty \quad \forall M \in \mathbb{R}, \exists a \in \mathbb{R} \text{ t.c. } \frac{x^2}{x+1} > M \quad \forall x > a$$

$$\frac{x^2 - Mx - M}{x+1} > 0 \quad x^2 - Mx - M > 0 \quad x \rightarrow 0 \text{ È DEFINITIVAMENTE } > 0 \quad (x > a_1)$$

$$x+1 > 0 \quad x > -1$$

$$\text{QUINDI } \frac{x^2}{x+1} > M \quad \forall x > \underbrace{\max\{-1, a_1\}}_a$$

LIMITI DESTRO E SINISTRO:

DEF: $f: A \rightarrow \mathbb{R}$, $x_0 \in \overline{\mathbb{R}}$ PUNTO DI ACCUM. DI A , $l \in \mathbb{R}$: $\lim_{\substack{x \rightarrow x_0^+ \\ (x \rightarrow x_0^-)}} f(x) = l$ SE

$\forall V$ INTORNO DI l , $\exists \delta > 0$ t.c. $f(x) \in V \quad \forall x$ t.c. $0 < x - x_0 < \delta$
($-\delta < x - x_0 < 0$)

$$\bullet \lim_{x \rightarrow 0^-} e^{\frac{1}{x}} = 0 \quad \text{OVERO } \forall \varepsilon > 0, \exists \delta > 0 \text{ t.c. } |e^{\frac{1}{x}}| < \varepsilon \quad \forall x \text{ t.c. } -\delta < x < 0$$

$$e^{\frac{1}{x}} < \varepsilon \Leftrightarrow \frac{1}{x} < \log \varepsilon \Leftrightarrow \frac{x < 0}{\varepsilon < 1} \quad \frac{1}{\log \varepsilon} < x < 0 \quad \delta = -\frac{1}{\log \varepsilon}$$

$$\lim_{x \rightarrow 0^+} e^{\frac{1}{x}} = +\infty \quad \text{OVERO } \forall M \in \mathbb{R}, \exists \delta > 0 \text{ t.c. } e^{\frac{1}{x}} > M \quad \forall x \text{ t.c. } 0 < x < \delta$$

$$e^{\frac{1}{x}} > M \Leftrightarrow \frac{1}{x} > \log M \Leftrightarrow \frac{x > 0}{M > 1} \quad 0 < x < \frac{1}{\log M} \quad \delta = \frac{1}{\log M}$$

TEO (CONFRONTO): $f, g: A \rightarrow \mathbb{R}$, $A \subseteq \mathbb{R}$, x_0 PTO DI ACC. DI A , $l_1, l_2 \in \overline{\mathbb{R}}$:

1) SE $f(x) \leq g(x)$ DEFINITIVAMENTE PER $x \rightarrow x_0$

2) $\exists \lim_{x \rightarrow x_0} f(x) = l_1 \quad \exists \lim_{x \rightarrow x_0} g(x) = l_2$

ALLORA: $l_1 \leq l_2$

TEO (2 CARABINIERI): $A \subseteq \mathbb{R}$, $f, g, h: A \rightarrow \mathbb{R}$, x_0 PUNTO DI ACCUMULAZIONE DI A , $l \in \mathbb{R}$:

1) SE $f(x) \leq g(x) \leq h(x)$ DEFINITIVAMENTE PER $x \rightarrow x_0$

2) $\exists \lim_{x \rightarrow x_0} f(x) = l \wedge \exists \lim_{x \rightarrow x_0} h(x) = l$

ALLORA: $\lim_{x \rightarrow x_0} g(x) = l$

DIM: $\forall \varepsilon > 0, \exists V_1, V_2$ INTORNI DI x_0 t.c.

$$x \in (A \cap V_1) \setminus \{x_0\} \Rightarrow |f(x) - l| < \varepsilon \Leftrightarrow l - \varepsilon < f(x) < l + \varepsilon$$

$$x \in (A \cap V_2) \setminus \{x_0\} \Rightarrow |h(x) - l| < \varepsilon \Leftrightarrow l - \varepsilon < h(x) < l + \varepsilon$$

$$\exists V_3 \text{ INTORNO DI } x_0 \text{ t.c. } x \in (A \cap V_3) \setminus \{x_0\} \quad f(x) \leq g(x) \leq h(x)$$

QUINDI $U = V_1 \cap V_2 \cap V_3$ INTORNO DI x_0 ALLORA:

$$\text{SE } x \in (U \cap A) \setminus \{x_0\} \Rightarrow l - \varepsilon < f(x) \leq g(x) \leq h(x) < l + \varepsilon \quad \text{CIOE'}$$

$$l - \varepsilon < g(x) < l + \varepsilon \Leftrightarrow |g(x) - l| < \varepsilon$$

TEO (SINGOLO CAPOBINIENE): $A \subseteq \mathbb{R}$, $f, g: A \rightarrow \mathbb{R}$, x_0 PTO DI ACCUMULAZIONE DI A , SE $f(x) \leq g(x)$

DEFINITIVAMENTE PER $x \rightarrow x_0$ ALLORA:

$$1) \text{ SE } \lim_{x \rightarrow x_0} f(x) = +\infty \Rightarrow \lim_{x \rightarrow x_0} g(x) = +\infty \quad 2) \text{ SE } \lim_{x \rightarrow x_0} g(x) = -\infty \Rightarrow \lim_{x \rightarrow x_0} f(x) = -\infty$$

TEO (LIMITE DELLA SOMMA CASO-FINITO): $A \subseteq \mathbb{R}$, $f, g: A \rightarrow \mathbb{R}$, $x_0 \in \mathbb{R}$ PTO DI ACCUMULAZIONE DI A :

SE $\lim_{x \rightarrow x_0} f(x) = l_1 \wedge \lim_{x \rightarrow x_0} g(x) = l_2$ ALLORA:

$$\lim_{x \rightarrow x_0} (f(x) + g(x)) = \lim_{x \rightarrow x_0} f(x) + \lim_{x \rightarrow x_0} g(x) = l_1 + l_2$$

NM: $\forall \varepsilon > 0 \quad \exists V_1, V_2$ INTORNI DI x_0 t.c.: POSSO SCEGLIERE QUALUNQUE ε

$$x \in (V_1 \cap A) \setminus \{x_0\} \Rightarrow |f(x) - l_1| < \frac{\varepsilon}{2} \Leftrightarrow l_1 - \frac{\varepsilon}{2} < f(x) < l_1 + \frac{\varepsilon}{2}$$

$$x \in (V_2 \cap A) \setminus \{x_0\} \Rightarrow |g(x) - l_2| < \frac{\varepsilon}{2} \Leftrightarrow l_2 - \frac{\varepsilon}{2} < g(x) < l_2 + \frac{\varepsilon}{2}$$

LI SOMMIAMO: SE $x \in (V_1 \cap V_2 \cap A) \setminus \{x_0\}$:

$$l_1 + l_2 - \frac{\varepsilon}{2} - \frac{\varepsilon}{2} < f(x) + g(x) < l_1 + l_2 + \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \Leftrightarrow$$

$$(l_1 + l_2) - \varepsilon < f(x) + g(x) < (l_1 + l_2) + \varepsilon \Leftrightarrow |f(x) + g(x) - (l_1 + l_2)| < \varepsilon$$